SOIL EROSION AND CONSERVATION · NOTE CM07

Vegetative Measures — Vegetative Barriers, Alley Cropping, Puerto Rican Terraces, Ley Farming and Mulching

Course topics: SWCE4.5 L8, SWCE7.4 L10, SWCE-510 T9

Vegetative barriers (live bunds, grass barriers, live hedges)

- Rows of vegetation along the contour that control erosion on farmland. Biological measures are eco-friendly and cheap, give fodder and thatch, and complement mechanical measures. Alone they suffice on milder slopes with little runoff and sediment; on steeper land with more runoff, combine them with mechanical measures.
- **How they work:** absorb raindrop energy, lengthen infiltration opportunity time, break slope length, add roughness, slow runoff and lower the peak, and make sediment settle on their upstream side. Plant material cushions raindrops; decomposing biomass improves structure and infiltration; roots bind particles and stabilise aggregates.
- **Grasses** establish quickly, with fibrous roots and dense tillers that slow runoff and anchor firmly; continuous root growth improves structure, fertility, biological activity and productivity.
- **Screen species** for regional adaptability, habitat, survival and growth, root and foliage traits, regeneration, and seed production and viability.

Environment	Species must…
Arid and semi-arid	Survive moisture deficit and frequent grazing; sprout new tillers quickly after the first rain; form an erect, stiff, dense hedge; bind soil; give economic products; occupy little width; not compete with crops, host pests or diseases, or become weeds
Flood plains	All the above, plus deep anchoring roots, tolerance of both waterlogging and drought, and recovery after inundation
Mountains and hill slopes	Dense, spreading roots with good lateral spread and a thick root mat at the surface; good seed production; frost and cold resistance

Region	Effective grasses
Shivaliks	Bhabar (*Eulaliopsis binata*), kans (*Saccharum spontaneum*)
Vertisols	*Cenchrus ciliaris*
Dry Alfisols	*Pennisetum hohenackeri*
Sub-humid lower western Himalayas	*Panicum maximum*
North-Eastern hills	*Cenchrus ciliaris*, *Panicum antidotale*, *Pennisetum polystachyon*
Doon valley (sub-humid)	Guinea grass most effective

Vetiver (*Vetiveria zizanioides*, *V. lawsonii*) has proved less effective in some regions — clump mortality, fire hazard, grazing, gaps and rhizome extraction — and failed in dry semi-arid Vertisols and Alfisols; it does better on sandy and clay loams and poorly drained land with a high water table.

Perennial grass	Notes
Cenchrus ciliaris	Drought tolerant; good hay
Cenchrus setigerus	Hardy, drought resistant
Apluda mutica	Medium fodder quality
Chrysopogon fulvus	Excellent tillering; degraded lands
Chloris gayana	Drought resistant; good on sandy loam
Brachiaria mutica	Humid climates; tolerates waterlogging
Cynodon plectostachyus	Rapid ground cover
Andropogon gayanus	Drought tolerant; up to 5 t/ha
Dichanthium annulatum	Heavy clay soils
Heteropogon contortus	Good hay
Lasiurus sindicus	Stabilises rolling sands
Panicum maximum	High biomass; bund protection
Panicum antidotale	Drought tolerant; green fodder
Pennisetum pedicellatum	Annual; drought resistant
Pennisetum polystachyon	Perennial; rainfed
Pennisetum purpureum	High fodder yield for 6–8 years
Sehima nervosum	Good-quality fodder
Thysanolaena maxima	Broom grass; moist hill slopes
Vetiveria zizanioides	Sandy loam soils

Problem land	Barrier species
Marginal land, deserts, sand dunes, ravines	*Cenchrus ciliaris*, *C. setigerus*, *Chrysopogon fulvus*, *Lasiurus sindicus*, *Atylosia scarabaeoides*, *Lablab purpureus*
Cultivable wastelands	*Cenchrus ciliaris*, *Sehima nervosum*, *Dichanthium annulatum*, *Chrysopogon fulvus*, *Stylosanthes* sp., *Clitoria ternatea*
Saline and alkaline lands	*Chloris gayana*, *C. barbata*, *Diplachne fusca*, *Lasiurus sindicus*, *Setaria sphacelata*, *Macroptilium atropurpureum*
Acid soils	*Pennisetum polystachyon*, *Paspalum notatum*, *Centrosema pubescens*, *Desmodium intortum*, *Stylosanthes guyanensis*
Marshy lands	*Brachiaria mutica*, *B. brizantha*
Vegetated waterways	Best: *Panicum repens*, *Brachiaria mutica*; also *Cynodon plectostachyus*, *C. dactylon*, *Paspalum notatum*, *Thysanolaena maxima*

- **Annuals** have simple roots and every ascending stem ends in an inflorescence; **perennials** have enduring roots (woody rootstock, rhizomes or stolons) and many shoots — essential for grass barriers.

- **Planting on mild slopes (< 6%):** form contour ditches with a V-ditcher or double mouldboard plough, heap the soil downstream as a small bund, and plant two slips in the ditch or on the bund; the ditch fills with soil over time and holds moisture.
- **Spacing:** vertical interval ≈ **50% of the recommended contour-bund spacing**, because bunds stop flow completely while hedges are permeable; horizontal spacing usually 20–40 m. Sediment builds up on the upstream side, so the barrier behaves as a dynamic living bund.
- Composite barriers combine trees and grasses — woody perennials such as *Gliricidia maculata* and *Leucaena leucocephala*, whose prunings are used as mulch against splash, to conserve moisture and add nutrients; prune trees that compete with crops.
- Barriers take 2–3 years to establish fully, are much cheaper than mechanical barriers, work better with a small downstream bund, are easy to establish and maintain, and give fodder and other produce.

Alley cropping

- An agroforestry system: food crops grow in alleys between closely planted, frequently pruned hedgerows of trees or shrubs at regular intervals. It suits marginal land, erratic rainfall, and soils of low fertility, high erodibility or shallow depth, on slopes up to 30%.
- Established hedges filter and hold soil, build natural terraces, reduce the erosivity of rain and runoff, dissipate energy, obstruct overland flow, bind soil and improve its physical and chemical properties. Prune during the crop season to reduce competition; establishment takes 2–3 years.
- **Alley width** varies with species, location and climate — commonly 2 to 10 m; wider alleys affect crops less.
- **Dehradun study:** contour paired rows of *Leucaena leucocephala* and eucalyptus with a 75 cm wide grass barrier at 1 m interval in maize cut runoff from 40% to 30% and soil loss from 21 to 8 t/ha.
- **Hedge species:** *Leucaena leucocephala* (nitrogen-fixing, soil-binding, improves soil) is most popular; also *Gliricidia sepium*, *Sesbania grandiflora*, *Robinia pseudoacacia* and *Pongamia pinnata*. They must tolerate lopping, grow fast and serve several uses.

Puerto Rican bench terraces

Mechanical or vegetative barriers along graded contours at suitable vertical intervals; soil moving downslope builds benches over 3–4 years. Mechanical barriers are bunds 30–40 cm high; vegetative barriers are tall, quick-growing grasses planted about 30 cm apart along graded lines, on soils with fair to good infiltration. No large-scale earthmoving, so cheaper than conventional bench terraces. Grasses: Guatemala grass (*Tripsacum laxum*), guinea grass (*Panicum maximum*), bhabbar (*Eulaliopsis binata*), hybrid napier, khus (*Vetiveria zizanioides*) and munj (*Saccharum munja*).

Ley farming

A crop rotation that includes a pasture crop (ley) to build fertility without chemical fertilisers — for dryland agriculture. Grass improves soil structure and legumes add nitrogen. Growing *Stylosanthes hamata* for 2–3 years in a 4-year rotation built soil nitrogen by up to 35 kg N/ha, keeping productivity without nitrogen fertiliser; farmer acceptance is not yet established.

Mulching and residue management

- Covering the soil between crop rows with residues, manures or litter reduces runoff and evaporation, induces infiltration and raises yields. In hills, FYM and leaf litter conserve moisture for ginger and

turmeric; in sugarcane, mulch lowers soil temperature and conserves moisture. Residues used: sorghum stubble, dry grass, wheat straw, pineapple stalks; tillage to form a dust mulch is also recommended.

- It is not widespread because vegetative matter is scarce in the dry season. Alternatives: harvested weeds, prunings, straw of rice, ragi and maize, toppings of sesbania, gliricidia, lantana, eucalyptus and *Acacia auriculiformis*.

Type	Practice
Live mulching	A mixed crop: a fast-growing, low, non-climbing legume that does not compete for light, moisture or nutrients, grown with a widely spaced grain crop (e.g. maize) and ploughed in at the right stage. Recycling sunnhemp in situ reduced erosion and raised the yield of the following wheat through better moisture and nutrition; dhaincha and cowpea are also promising as green-manure-cum-live mulch
Vertical mulching	Trenches 15 cm wide and 30 cm deep across the slope, packed with sorghum, maize or pearl-millet stubble protruding 10 cm above ground, 4–8 m apart (continuous or staggered). They act as open ditches that admit runoff and recharge the subsoil — effective on mild slopes in deep black soils, at 5 m intervals in central and south India
Dust mulching	Keeping the surface loose and friable by tillage, which slows moisture loss from the subsoil, controls weeds and conserves moisture and nutrients

Species names are given in their standard spelling (e.g. Centrosema for "Cenisrema", Stylosanthes hamata for "hamatai").

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. ICAR–Indian Institute of Soil and Water Conservation, Dehradun (vegetative barriers and alley cropping trials). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Vertical interval of vegetative barriers is about ___ of contour-bund spacing

(a) 25% (b) 50%
(c) 100% (d) 200%

Q2. Horizontal spacing of vegetative barriers is usually

(a) 2–5 m (b) 20–40 m
(c) 80–100 m (d) 200 m

Q3. Alley cropping is advantageous on slopes up to

(a) 5% (b) 15%
(c) 30% (d) 60%

Q4. In the Dehradun alley-cropping study, soil loss fell from 21 to

(a) 2 t/ha (b) 8 t/ha
(c) 15 t/ha (d) 18 t/ha

Q5. Puerto Rican terraces form benches over about

(a) 1 year (b) 3–4 years
(c) 10 years (d) 20 years

Q6. Ley farming with Stylosanthes hamata built soil N by up to

(a) 5 kg/ha (b) 35 kg/ha
(c) 100 kg/ha (d) 200 kg/ha

Q7. Vertical-mulch trenches are

(a) 15 cm wide, 30 cm deep
(b) 45 cm wide, 45 cm deep
(c) 1 m wide, 1 m deep
(d) 5 cm wide, 10 cm deep

Q8. Vetiver failed mainly in

(a) high water-table areas
(b) dry semi-arid Vertisols and Alfisols
(c) sandy loams
(d) clay loams

Q9. A grass best suited to stabilise rolling sands is

(a) Brachiaria mutica
(b) Lasiurus sindicus
(c) Paspalum notatum
(d) Thysanolaena maxima

Answers

Q1 — (b) 50% — *Hedges are permeable.*
Q2 — (b) 20–40 m
Q3 — (c) 30%
Q4 — (b) 8 t/ha
Q5 — (b) 3–4 years
Q6 — (b) 35 kg/ha
Q7 — (a) 15 cm wide, 30 cm deep
Q8 — (b) dry semi-arid Vertisols and Alfisols
Q9 — (b) Lasiurus sindicus